

TP 5.2 - Création d'un serveur Apache

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id: srvapache & mdp: apache

Exercice 1 — Installer Apache + vérifier

installer apache:

- `sudo apt update`
- `sudo apt install -y apache2`

```
srvapache@apache: ~  
srvapache@apache:~$ sudo apt install -y apache2  
[sudo] password for srvapache:  
Lecture des listes de paquets... Fait  
Construction de l'arbre des dépendances... Fait  
Lecture des informations d'état... Fait  
Les paquets supplémentaires suivants seront installés :  
  apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1t64 liblua5.4-0 ssl-cert  
Paquets suggérés :
```

vérifier:

- `sudo systemctl status apache2 --no-pager`

```
srvapache@apache:~$ sudo systemctl status apache2 --no-pager  
● apache2.service - The Apache HTTP Server  
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)  
   Active: active (running) since Thu 2026-01-29 13:38:27 UTC; 22s ago  
     Docs: https://httpd.apache.org/docs/2.4/  
  Main PID: 2537 (apache2)  
    Tasks: 55 (Limit: 4487)  
  Memory: 5.2M (peak: 5.3M)  
     CPU: 63ms  
    CGroup: /system.slice/apache2.service  
            └─2537 /usr/sbin/apache2 -k start  
              └─2539 /usr/sbin/apache2 -k start  
                └─2540 /usr/sbin/apache2 -k start  
  
janv. 29 13:38:26 apache systemd[1]: Starting apache2.service - The Apache HTTP Server...  
janv. 29 13:38:27 apache apachectl[2536]: AH00558: apache2: Could not reliably determine the se...  
janv. 29 13:38:27 apache systemd[1]: Started apache2.service - The Apache HTTP Server.  
Hint: Some lines were ellipsized, use -l to show in full.  
srvapache@apache:~$
```

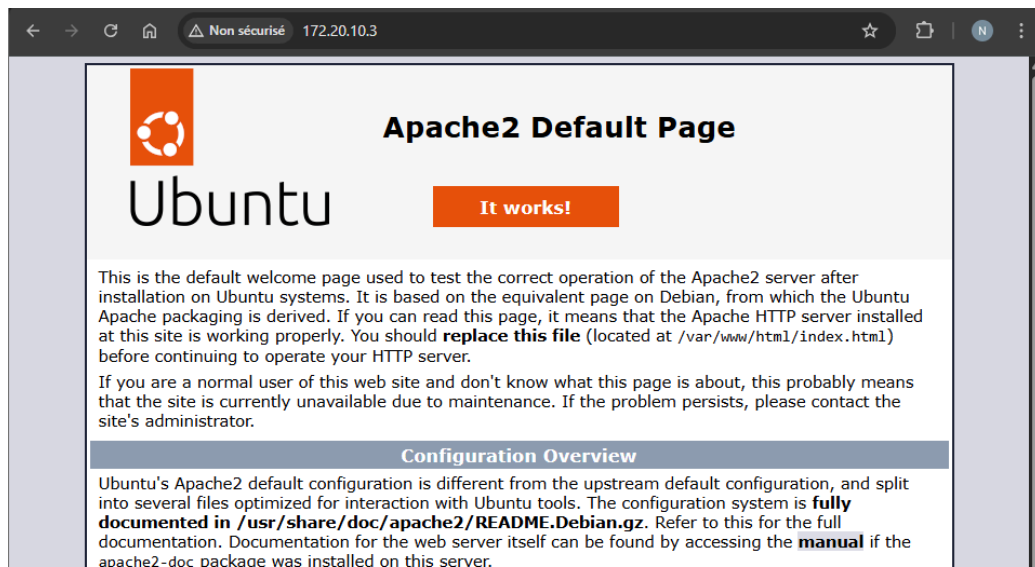
test:

- `curl -I http://localhost`
- `http://172.20.10.3`

sur la vm:

```
srvapache@apache:~$ curl -I http://localhost  
HTTP/1.1 200 OK  
Date: Thu, 29 Jan 2026 13:39:08 GMT  
Server: Apache/2.4.58 (Ubuntu)  
Last-Modified: Thu, 29 Jan 2026 13:38:24 GMT  
ETag: "29af-64986f88ea657"  
Accept-Ranges: bytes  
Content-Length: 10671  
Vary: Accept-Encoding  
Content-Type: text/html
```

sur le net:



Exercice 2 — Hébergement de plusieurs sites web

Créer l'arborescence:

- `sudo mkdir -p /var/www/site1/public_html`
- `sudo mkdir -p /var/www/site2/public_html`
- `sudo chown -R $USER:$USER /var/www/site1 /var/www/site2`

```
srvapache@apache:~$ sudo mkdir -p /var/www/site1/public_html
srvapache@apache:~$ sudo mkdir -p /var/www/site2/public_html
srvapache@apache:~$ sudo chown -R $USER:$USER /var/www/site1 /var/www/site2
srvapache@apache:~$
```

Créer des pages de test:

- `echo "<h1>Bienvenue sur SITE 1</h1>" > /var/www/site1/public_html/index.html`
- `echo "<h1>Bienvenue sur SITE 2</h1>" > /var/www/site2/public_html/index.html`

```
srvapache@apache:~$ echo "<h1>Bienvenue sur SITE 1</h1>" > /var/www/site1/public_html/index.html
srvapache@apache:~$ echo "<h1>Bienvenue sur SITE 2</h1>" > /var/www/site2/public_html/index.html
srvapache@apache:~$
```

Créer les VirtualHosts:

VH 1:

- `sudo nano /etc/apache2/sites-available/site1.conf`

```
GNU nano 7.2 /etc/apache2/sites-available/site1.conf *
<VirtualHost *:80>
  ServerName site1.local
  DocumentRoot /var/www/site1/public_html

  <Directory /var/www/site1/public_html>
    Require all granted
  </Directory>

  ErrorLog ${APACHE_LOG_DIR}/site1_error.log
  CustomLog ${APACHE_LOG_DIR}/site1_access.log combined
</VirtualHost>
```

VH 2:

- `sudo nano /etc/apache2/sites-available/site2.conf:`

```

GNU nano 7.2 /etc/apache2/sites-available/site2.conf *
<VirtualHost *:80>
    ServerName site2.local
    DocumentRoot /var/www/site2/public_html

    <Directory /var/www/site2/public_html>
        Require all granted
    </Directory>

    ErrorLog ${APACHE_LOG_DIR}/site2_error.log
    CustomLog ${APACHE_LOG_DIR}/site2_access.log combined
</VirtualHost>

```

Activer les sites:

- sudo a2ensite site1.conf
- sudo a2ensite site2.conf
- sudo a2dissite 000-default.conf
- sudo apache2ctl configtest
- sudo systemctl reload apache2

```

srvapache@apache:~$ sudo a2ensite site1.conf
Enabling site site1.
To activate the new configuration, you need to run:
systemctl reload apache2
srvapache@apache:~$ sudo a2ensite site2.conf
Enabling site site2.
To activate the new configuration, you need to run:
systemctl reload apache2
srvapache@apache:~$ sudo a2dissite 000-default.conf
Site 000-default disabled.
To activate the new configuration, you need to run:
systemctl reload apache2
srvapache@apache:~$ sudo apache2ctl configtest
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1
.1. Set the 'ServerName' directive globally to suppress this message
Syntax OK
srvapache@apache:~$ sudo systemctl reload apache2
srvapache@apache:~$

```

Tester l'accès:

- curl -H "Host: site1.local" http://localhost
- curl -H "Host: site2.local" <http://localhost>

```

srvapache@apache:~$ curl -H "Host: site1.local" http://localhost
<h1>Bienvenue sur SITE 1</h1>
srvapache@apache:~$ curl -H "Host: site2.local" http://localhost
<h1>Bienvenue sur SITE 2</h1>
srvapache@apache:~$

```

Exercice 3 — Sécurisation

- curl -I <http://localhost>
- curl <http://localhost/404-test>

```
srvapache@apache:~$ curl -I http://localhost
HTTP/1.1 200 OK
Date: Thu, 29 Jan 2026 13:49:45 GMT
Server: Apache/2.4.58 (Ubuntu)
Last-Modified: Thu, 29 Jan 2026 13:42:14 GMT
ETag: "1e-64987063bfe45"
Accept-Ranges: bytes
Content-Length: 30
Content-Type: text/html

srvapache@apache:~$ curl http://localhost/404-test
<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>404 Not Found</title>
</head><body>
<h1>Not Found</h1>
<p>The requested URL was not found on this server.</p>
<hr>
<address>Apache/2.4.58 (Ubuntu) Server at localhost Port 80</address>
</body></html>
srvapache@apache:~$ |
```

Avec ça on voit le logiciel, la version, l'OS, c'est pas sécurisé on a accès à beaucoup trop d'informations .

Appliquer durcissement:

- sudo nano /etc/apache2/conf-available/security.conf
- sudo systemctl reload apache2

avant:

```
#ServerTokens Minimal
ServerTokens OS
#ServerTokens Full

#
# Optionally add a line containing the server version and virtual host
# name to server-generated pages (internal error documents, FTP directory
# listings, mod_status and mod_info output etc., but not CGI generated
# documents or custom error documents).
# Set to "EMail" to also include a mailto: link to the ServerAdmin.
# Set to one of: On | Off | EMail
#ServerSignature Off
ServerSignature On

#
# Allow TRACE method
#
# Set to "extended" to also reflect the request body (only for testing and
# diagnostic purposes).
#
# Set to one of: On | Off | extended
TraceEnable Off
#TraceEnable On
```

apres:

```

ServerTokens Prod
#ServerTokens Full

#
# Optionally add a line containing the server version and virtual host
# name to server-generated pages (internal error documents, FTP directory
# listings, mod_status and mod_info output etc., but not CGI generated
# documents or custom error documents).
# Set to "Email" to also include a mailto: link to the ServerAdmin.
# Set to one of: On | Off | Email
#ServerSignature Off
ServerSignature Off

#
# Allow TRACE method
#
# Set to "extended" to also reflect the request body (only for testing and
# diagnostic purposes).
#
# Set to one of: On | Off | extended
TraceEnable Off
#TraceEnable On

```

ServerTokens Prod = ne dévoile pas OS/version précise
 ServerSignature Off = retire la signature dans pages d'erreur
 TraceEnable Off = désactive TRACE (réduit surface)

```

srvapache@apache:~$ sudo nano /etc/apache2/conf-available/security.conf
srvapache@apache:~$ sudo systemctl reload apache2
srvapache@apache:~$ |

```

```

srvapache@apache:~$ curl -I http://localhost
HTTP/1.1 200 OK
Date: Thu, 29 Jan 2026 13:55:06 GMT
Server: Apache
Last-Modified: Thu, 29 Jan 2026 13:42:14 GMT
ETag: "1e-64987063bfe45"
Accept-Ranges: bytes
Content-Length: 30
Content-Type: text/html

srvapache@apache:~$ |

```

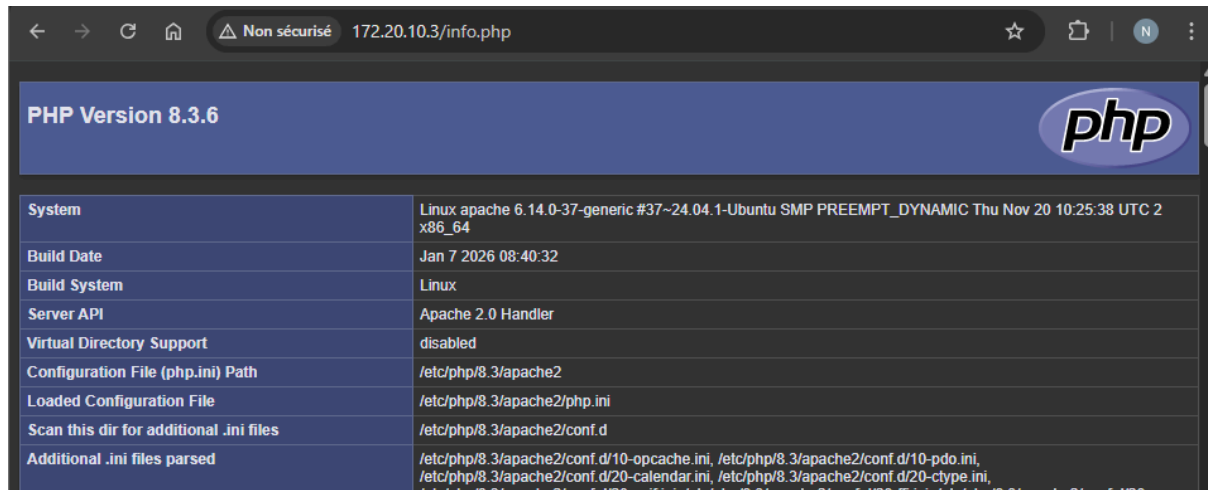
On voit ici que les données sensibles ne sont plus affichées.

Bonus 1:

- sudo apt install -y php libapache2-mod-php
- sudo systemctl reload apache2
- echo "<?php phpinfo(); ?>" | sudo tee /var/www/site1/public_html/info.php
- http://172.20.10.3/info.php
- sudo rm /var/www/site1/public_html/info.php

```
srvapache@apache:~$ sudo apt install -y php libapache2-mod-php
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
Les paquets supplémentaires suivants seront installés :
  libapache2-mod-php8.3 php-common php8.3 php8.3-cli php8.3-common php8.3-opcache php8.3-xml
Paquets suggérés :
  libapache2-ssl-modules php-pear
srvapache@apache:~$ sudo systemctl reload apache2
srvapache@apache:~$ echo "<?php phpinfo(); ?>" | sudo tee /var/www/site1/public_html/info.php
<?php phpinfo(); ?>
srvapache@apache:~$
```

on ouvre l'interface graphique:



on supprime le fichier sensible:

```
srvapache@apache:~$ sudo find /var/www -name info.php
*/var/www/site1/public_html/info.php
srvapache@apache:~$ rm /var/www/site1/public_html/info.php
rm: remove write-protected regular file '/var/www/site1/public_html/info.php'?
srvapache@apache:~$
```

vérification:

- ls /var/www/site1/public_html

```
srvapache@apache:~$ ls /var/www/site1/public_html
index.html
srvapache@apache:~$
```



```
Group: name="www-data" id=33  
srvapache@apache:~$ curl -H "Host: site1.local" http://localhost  
<h1>Bienvenue sur SITE 1</h1>  
srvapache@apache:~$
```

Après vérification du fonctionnement de PHP via la page info.php, celle-ci a été supprimée afin d'éviter l'exposition d'informations sensibles sur la configuration du serveur.